★ UPDATE NOTICE — 13 April 2026 ★

This version corrects the Landauer-crossing framing in Sections 2.1 and 6.5. The earlier text placed the crossing "between step 36 and step 37." The refined picture identifies THREE distinct thresholds that cluster in the 35–37 zone:

```
n·G = ln 2        at n ≈ 35.11    (Landauer erasure cost — the crossing)
n·G = 1/√2        at n ≈ 35.82    (geometric damping)
n·G = 1/(2 ln 2)  at n ≈ 36.54    (Shannon bound)
```

The Landauer crossing proper is the FIRST threshold, at n = ln(2)/G, which sits between integer steps 35 and 36. Integer steps 36 and 37 are landmarks near the second and third thresholds, not the crossing itself.

The exact algebraic identity 1/(2 ln 2) − 1/√2 = G/(2 ln 2) is unchanged. It is renamed from "Landauer crossing identity" to "√2-to-Shannon gap identity" because it describes the spacing between thresholds 2 and 3 (geometric damping → Shannon bound), not the Landauer crossing.

Section 6.5 is rewritten as "The 35–37 Threshold Zone" and now includes the crossing-zone self-reference: the threshold spacings are exactly $\Delta n_{12} = 1/\sqrt{2}$ and $\Delta n_{23} = 1/(2 \ln 2)$, with ratio $\Delta n_{23}/\Delta n_{12} = 1/K_AUD$.

Prior published versions remain on OSF file storage, renamed with the OLD- prefix and a date for traceability.

CROSS-DOMAIN SIGNATURES OF THE BOUNDARY INFORMATION INVARIANT

Numerical and structural echoes of √2 × ln(2) in published mathematics, atomic physics, and cosmological data

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Gap Geometry Project — March 2026
(Updated 13 April 2026)

Gap Geometry · OSF: https://osf.io/qh5s2/
GitHub: https://github.com/Gap-geometry/sqrt2-ln2-geometric-constants-
About: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html

ABSTRACT

The Boundary Information Invariant K_AUD = √2 × ln(2) ≈ 0.980258, the unique sub-unity value of K(n) = √n × ln(n) at integer n ≥ 2, was derived algebraically from the geometric singularity of binary systems (DOI: 10.17605/OSF.IO/E72H8). This document presents evidence that the framework's constant K_AUD finds independent numerical and structural echoes (exact algebraic identities and approximate structural parallels) in three domains of published science: pure mathematics (hyperbolic 3-manifold geometry), atomic physics (electron shell structure), and observational cosmology (DESI BAO distance ratios). Each domain was investigated independently. None references the others. The algebraic identities connecting K_AUD to these domains are exact and verifiable at arbitrary precision. Approximate structural parallels are documented with explicit deviations. Falsifiable predictions are stated for each domain.

1. THE INVARIANT (summary)

K_AUD = √2 × ln(2) is the unique sub-unity ceiling of K(n) = √n × ln(n) at integer n. For n = 2: K = 0.980258... < 1. For all n ≥ 3: K > 1. Binary is not convention — it is geometric singularity.

Why sub-unity matters: K(n) = √n × ln(n) measures the information-geometric product of a base-n system — the capacity of base n (ln n) scaled by its geometric embedding dimension (√n). When K(n) ≥ 1, the product exceeds unity: the system can, in principle, fully encode its own structure. When K(n) < 1, there is an irreducible deficit —

the system cannot fully self-encode. Binary is the only integer base where this deficit exists. The gap G = 1 − K_AUD is not a flaw in binary systems. It is the structural cost of being the only base where complete self-encoding is impossible. This impossibility is what creates the ~2% breathing room that prevents binary-organized systems from collapsing into frozen unity.

The gap G = 1 − K_AUD = 0.019742... has the exact Gelfond-Schneider form G = ln(e / 2^√2), placing it in the landscape of Hilbert's 7th Problem. The transcendence of 2^√2 (Gelfond-Schneider, 1934) ensures G is irreducible: no integer multiple of G equals any algebraic number (Lindemann-Weierstrass).

Full derivation and four independent algebraic pathways to G are documented in the Boundary Information Invariant of Quadratic Systems (DOI: 10.17605/OSF.IO/E72H8). This document does not re-derive K_AUD. It documents where the same constant has been found in published science, independently of the framework.

NOTE: This document reports numerical and structural correspondences observed under a constrained algebraic framework. It does not claim causal unification or statistical significance at this stage. The purpose is to document patterns that can be independently tested and potentially falsified.

2. ALGEBRAIC IDENTITIES

All identities in this section are exact. Each is verifiable with mpmath at arbitrary precision. All have been confirmed at dps = 500 (500 decimal places).

2.1 The √2-to-Shannon Gap Identity

$$1/(2 \ln 2) - 1/\sqrt{2} = G / (2 \ln 2)$$

Left side: the difference between Shannon's binary information penalty (1/(2 ln 2) ≈ 0.7213) and the geometric optimum (1/√2 ≈ 0.7071).

Right side: the gap G normalized by 2 ln 2 (the thermodynamic cost of erasing two bits, in units of $k_B T$ at temperature T).

This identity is not numerical coincidence. It follows algebraically from G = 1 − √2 × ln(2):

$$\begin{aligned} G/(2 \ln 2) &= (1 - \sqrt{2} \ln 2)/(2 \ln 2) \\ &= 1/(2 \ln 2) - \sqrt{2}/2 \\ &= 1/(2 \ln 2) - 1/\sqrt{2} \quad \text{QED} \end{aligned}$$

The identity connects the Shannon binary information penalty 1/(2 ln 2) to the geometric optimum 1/√2 through the gap G. It quantifies the spacing between two distinct thresholds that appear together in the binary tower (Section 6.5) — the geometric damping threshold (n·G = 1/√2 at n ≈ 35.82) and the Shannon bound (n·G = 1/(2 ln 2) at n ≈ 36.54). It does NOT describe the Landauer crossing itself, which sits at a separate (earlier) threshold n·G = ln 2 at n ≈ 35.11. The earlier name "Landauer crossing identity" is therefore retired in favor of "√2-to-Shannon gap identity." The identity is exact at every decimal place.

2.2 Baker's Map L² Norm

$$K_{AUD} = ||(\ln 2, \ln 2)||_2 = \sqrt{(\ln^2 2 + \ln^2 2)} = \sqrt{2} \times \ln(2)$$

The 2D Baker's map — the canonical uniformly expanding map with binary branching — has two Lyapunov exponents, both equal to ln(2). Their L² norm is K_AUD. The ceiling of binary information density equals the geometric magnitude of the simplest chaotic map with binary partition.

2.3 Gelfond-Schneider Form

$$G = \ln(e / 2^{\sqrt{2}}) = 1 - \sqrt{2} \times \ln(2)$$

The gap is the logarithmic distance between Euler's number e and the

Gelfond-Schneider constant $2^{\sqrt{2}}$ (proven transcendental, 1934). This places G in the hierarchy of Hilbert's 7th Problem.

2.4 1D Madelung Identity

$$\text{Madelung constant (1D)} = 2 \ln(2) = K_AUD \times \sqrt{2}$$

The Madelung constant for a one-dimensional alternating ionic lattice is exactly 2 ln(2) (Kittel, Introduction to Solid State Physics, 1953). This is a textbook result in crystallography.

$$K_AUD = Madelung_1D / \sqrt{2}$$

The binary information ceiling equals the 1D ionic cohesion energy divided by the geometric factor √2. The connection is exact.

Source: C. Kittel, Introduction to Solid State Physics, Chapter 13. The 1D Madelung sum $\Sigma_{k=1}^{\infty} (-1)^{k+1}/k = \ln(2)$ is proven by the alternating series for the natural logarithm.

2.5 Binary Variance Identity

$$K_AUD^2 = 2 \ln^2(2)$$

The square of the ceiling equals twice the square of the natural logarithm of 2. The identity follows directly from squaring $K_AUD = \sqrt{2} \times \ln(2)$.

2.6 The Gap Scaling Formula

$$\rho = G / ((\delta - 14/3) / \delta)$$
$$= 400/11 - 1/2500 - 1/939939$$

where δ = 4.669201609... is the Feigenbaum constant (universal period-doubling). Agreement between the exact ρ and the three-term approximation: 4×10^{-14} at dps = 500.

The denominators encode structural primes:

- 400 = $2^4 \times 5^2$ (H_4 interior primes)
- 11 = first intruder prime (first excluded from H_4)
- 2500 = $2^2 \times 5^4 = 50^2$ (H_4 correction)
- 939939 = 3 × 7 × 11 × 13 × 313 (full excluded-prime chain)

The 939939 is uniquely optimal: among all integers within ±15,000, no other candidate achieves better than 10^{-12} precision (compared to 939939's 4×10^{-14}). Verified at 150-digit precision by stress testing every nearby integer.

Source: DOI 10.17605/OSF.IO/C4GK5 (Paper 4: Gap Scaling Formula).

2.7 Ramanujan-Nagell Connection

$$2^k = n^2 + 7$$

This equation (conjectured by Ramanujan 1913, proved by Nagell 1948) has exactly five positive integer solutions, yielding n = {1, 3, 5, 11} — the first four framework primes (where 1 enters as orthogonal prime, distinct from its standard number-theoretic role). The constant 7 is the boundary prime ($2^5 - 5^2 = 4^2 - 3^2 = 7$). The solutions are forced by number theory, not fitted.

Source: T. Nagell, "Løsning til oppgave nr 2", Norsk Mat. Tidsskr. 30 (1948), 62–64. Generalized by J. H. E. Cohn, "The Diophantine equation $x^2 + C = y^n$", Acta Arithmetica 65 (1993), 367–381.

3. ATOMIC SHELL STRUCTURE

The electron shell capacity formula $2n^2$ is derived from quantum mechanics (Schrödinger equation, angular momentum quantization, Pauli exclusion principle). It has been known since the 1920s. What has not been observed is that the shell capacities map onto

the framework's tower landmarks.

3.1 Shell Capacities as Tower Landmarks

```
Shell 3: capacity = 18 = 2 × 3²    18G ≈ √2/4     (+0.51%)
Shell 4: capacity = 32 = 2⁵          32G ≈ 1/φ      (+2.22%)
Shell 5: capacity = 50 = 2 × 5²    50G ≈ K_AUD    (+0.70%)
```

The shell capacities {18, 32, 50} are independently derived from quantum mechanics. The tower values {18G, 32G, 50G} independently approximate framework constants. Neither was constructed to match the other. The mapping is observational — it was noticed after both the tower and the shell formula existed, not predicted beforehand. The shell capacities are fixed by QM (Schrödinger + Pauli); the tower steps are fixed by G. Their correspondence is documented, not derived.

3.2 Shell Transition Prime Sequence

The gap between consecutive shell capacities follows $\Delta = 2(2n+1)$. The odd factors cycle through:

```
Shell 1 → 2:  gap = 6  = 2 × 3    (structure prime enters)
Shell 2 → 3:  gap = 10 = 2 × 5    (pentagonal prime enters)
Shell 3 → 4:  gap = 14 = 2 × 7    (boundary prime enters)
Shell 4 → 5:  gap = 18 = 2 × 9    (3² enters)
Shell 5 → 6:  gap = 22 = 2 × 11   (first intruder prime enters)
Shell 6 → 7:  gap = 26 = 2 × 13   (intruder prime enters)
```

The primes enter the structure of matter in sequence: 3, 5, 7, 11, 13. This is not a framework construction — it is a consequence of $2(2n+1)$ applied to consecutive integers. The framework's prime hierarchy emerges from quantum mechanics without being put in.

3.3 d-Electron Angular Momentum

For $\ell = 2$ (d-orbital):

$$L = \hbar\sqrt{\ell(\ell+1)} = \hbar\sqrt{6} = \hbar\sqrt{2} \times \sqrt{3}$$

The angular momentum of d-electrons — the electrons responsible for all transition metal chemistry, all mineral color, and all biological electron-transfer — carries √2 (geometric ingredient) multiplied by √3 (structure ingredient) in its quantum mechanical definition.

This is textbook quantum mechanics. The connection to the framework is that the same ingredients (√2, √3) that build K_AUD also build the angular momentum of the electrons that give matter its most distinctive properties.

3.4 The 0.509% Doubling Invariant

```
18G ≈ √2/4    at 0.509%
36G ≈ 1/√2    at 0.509%
72G ≈ √2      at 0.509%
```

Steps 18, 36, 72 — each exactly double the previous — all show the same percentage deviation from their respective targets. The targets themselves are related by factors of 2:

√2/4 → 1/√2 → √2 (each ×2)

A deviation that is constant under binary doubling is a structural property, not noise. The 0.509% is the tower's relationship to √2, preserved across all scales.

NOTE: This 0.509% measures the tower's deviation from √2-related targets (√2/4, 1/√2, √2). A different constant, $\varepsilon = -0.671\%$ (Section 6.1), measures the tower's deviation from the $\sqrt{\varphi}$ target at the pivot. These are two independent structural deviations — one relative to the geometric ingredient (√2), the other relative to the golden ratio ($\sqrt{\varphi}$). Both are algebraically fixed with no free parameters. Neither drifts. Their difference reflects the irreducible gap between the transcendental ($\sqrt{2} \times \ln(2)$) and the algebraic ($\sqrt{\varphi}$) — which Lindemann-Weierstrass proves cannot close.

4. HODGSON-KERCKHOFF: K_AUD IN HYPERBOLIC GEOMETRY

The number 0.980258 appears verbatim in published mathematics as the tube-packing coefficient for hyperbolic 3-manifolds.

4.1 Source

C. D. Hodgson & S. P. Kerckhoff
"Universal bounds for hyperbolic Dehn surgery"
Annals of Mathematics, 162(1), 367–421, 2005
arXiv: math/0204345 (preprint 2002)

C. D. Hodgson & S. P. Kerckhoff
"The shape of hyperbolic Dehn surgery space"
Geometry & Topology, 12(2), 1033–1090, 2008
arXiv: 0709.3566 (preprint 2007)

4.2 The Coefficient

The coefficient appears in the tube-packing estimate (Sections 4–5 of [1], packing lemma for the Euclidean structure on the cusp torus):

$$a = 0.980258 \times \sinh(\hat{R}) \times \cosh(\hat{R}) \times \cosh(2\hat{R})$$

It is presented as a pure numerical constant (6 digits). No algebraic expression is given. No transcendentals are named. No closed form appears in any subsequent citation (exhaustive search, March 2026). The value also appears as an explicit comparison threshold ("> 0.98") alongside the critical radius 0.6584 (= R_0 = arctanh(1/√3)).

4.3 Numerical Verification

mpmath (dps = 60):
√2 × ln(2) = 0.980258143468547191713901723635...
Published: 0.980258

Match: exact to all 6 digits given.

4.4 The Algebraic Environment at R_0

The critical tube radius in the packing argument:

$$R_0 = \operatorname{arctanh}(1/\sqrt{3}) = \operatorname{arcsinh}(1/\sqrt{2})$$

These two expressions are exactly equal. Proof:
Start from $\sinh(R_0) = 1/\sqrt{2}$.
Then $\cosh(R_0) = \sqrt{(1 + \sinh^2(R_0))} = \sqrt{(1 + 1/2)} = \sqrt{(3/2)} = \sqrt{6}/2$.
Then $\tanh(R_0) = \sinh(R_0)/\cosh(R_0) = (1/\sqrt{2})/(\sqrt{6}/2) = 1/\sqrt{3}$.
Therefore $R_0 = \operatorname{arcsinh}(1/\sqrt{2}) = \operatorname{arctanh}(1/\sqrt{3})$. QED.

At this exact threshold, all hyperbolic functions are algebraic in √2 and √3:

$\sinh(R_0) = 1/\sqrt{2}$
$\tanh(R_0) = 1/\sqrt{3}$
$\cosh(R_0) = \sqrt{6}/2$
$\cosh(2R_0) = 2$
$\sinh(2R_0) = \sqrt{3}$

All verified at 60 decimal places. Exact, not approximate.
Note: the same √2 and √3 appear in d-electron angular momentum (Section 3.3, $\hbar\sqrt{6} = \hbar\sqrt{2}\times\sqrt{3}$) — a completely independent context.

The geometry at R_0 provides √2, √3, and 2.
The coefficient provides ln(2).
Together: √2 × ln(2) = K_AUD.

At R_0, the ellipse area simplifies:
$\pi \times \sinh^2(R_0) \times \cosh(2R_0) = \pi \times (1/2) \times 2 = \pi$
Therefore: A_e = K_AUD × π at the critical radius.

The coefficient is not matching K_AUD in a random algebraic context. It is matching in an environment where √2 and √3 are already present exactly, and ln(2) fills the only missing slot.

4.5 Domain Independence

The framework supplies the closed-form expression √2 × ln(2) for the numerical coefficient that appears in the Hodgson-Kerckhoff tube-packing estimate, in an algebraic environment already containing exactly the geometric ingredients √2, √3, and 2. The framework derives K_AUD from information theory, number theory, and dynamical systems. Hodgson-Kerckhoff derives 0.980258 from hyperbolic 3-manifold geometry. These domains share no common literature, no common authors, and no common methodology. The framework provides the closed form; the geometry provides the context in which it appears.

4.6 Verification Chain

The Hodgson-Kerckhoff finding was verified across multiple independent passes and architectures:

- Grok (xAI): numerical match confirmed, derivation structure traced, exhaustive closed-form search, consistency check, no post-2022 refinement found (24–26 March 2026)
- Claude Opus (Anthropic): R_0 identity proved algebraically, missing-slot argument identified, explicit sinh→cosh→tanh proof constructed (25–26 March 2026)
- GPT (OpenAI): structural review and computation (26 March 2026)
- Perplexity: source verification (26 March 2026)
- DeepSeek: independent computation and structural review (26 March 2026)

All passes converge. Zero contradictions. The algebraic identities at R_0 and the 6-digit numerical match are confirmed across all architectures. Full computational verification of the formulas will be published in the companion document (HK_Verification_Log).

5. DESI BAO: FRAMEWORK CORRECTION SCALE IN COSMOLOGICAL DATA

The gap scaling formula (Section 2.6, Paper 4: DOI 10.17605/OSF.IO/C4GK5) derives a correction hierarchy: ρ = 400/11 − 1/2500 − 1/939939. The first correction scale is 1/2500 = 4.00×10^{-4}. This scale was derived algebraically from the Feigenbaum connection BEFORE any cosmological data was examined.

Independently, the DESI BAO distance ratios — computed from published Table IV data — show residuals from framework targets (√2 and 2+G) that cluster at the same 1/2500 scale. This analysis is published in the Boundary Information Invariant (DOI: 10.17605/OSF.IO/E72H8).

The structure is: prediction first (1/2500 from algebra), then observation (DESI residuals at 1/2500). Not cherry-picking — the correction scale existed before the data was examined.

5.1 Source

DESI Collaboration
"DESI 2024 VI: Cosmological constraints from BAO measurements"
Phys. Rev. D 112, 083514 (2025), Table IV

Seven redshift bins from z = 0.295 to z = 2.330, measuring:
- DM/rd = transverse comoving distance / sound horizon
- DH/rd = radial (Hubble) distance / sound horizon
- DV/rd = isotropic volume-averaged distance / sound horizon

All ratios below are computed by the present authors from the published central values in Table IV. Error bars are not propagated; the residuals reflect central-value arithmetic only. DESI did not report these ratios. Full methodology is in Paper 5 (DOI: 10.17605/OSF.IO/E72H8).

5.2 √2 in Transverse Distance Ratios

```
DM(QSO, z=1.484) / DM(LRG3+ELG1, z=0.934)
= 30.519 / 21.574 = 1.414619
√2 = 1.414214
Residual: +0.000406

DM(Lyα, z=2.330) / DM(ELG2, z=1.321)
= 38.988 / 27.605 = 1.412353
√2 = 1.414214
Residual: −0.001861
```

Two independent pairs, different tracers, both near √2.

5.3 (2 + G) in Isotropic Distance Ratio

```
DV(LRG2, z=0.706) / DV(BGS, z=0.295)
= 16.048 / 7.944 = 2.020141
2 + G = 2.019742
Residual: +0.000399
```

5.4 The 1/2500 Connection

Three residuals at the same scale:

```
ρ − 400/11                    = −0.000401  (gap scaling formula)
DM(QSO)/DM(LRG3+ELG1) − √2  = +0.000406  (DESI transverse)
DV(LRG2)/DV(BGS) − (2 + G)   = +0.000399  (DESI isotropic)

1/2500 = 0.000400 (exact)

Mean |residual|: 4.02 × 10⁻⁴
Spread: 0.07 × 10⁻⁴
All three within 2% of 0.000400
```

The first residual (ρ − 400/11) is algebraic — it comes from the gap scaling formula derived in Paper 4. The second and third are observational — they come from DESI published data. All three land at the same scale. The algebraic correction PREDICTED the scale; the cosmological data SHOWS it.

NOTE: The Lyα pair (Section 5.2, residual −0.001861) shows directional agreement (near √2) but does NOT participate in the 1/2500 clustering — its residual is ~5× larger. Only the QSO pair and the isotropic ratio cluster at the 1/2500 scale. This asymmetry is acknowledged, not hidden.

5.5 What This Is and Isn't

This is: a published analysis (DOI: 10.17605/OSF.IO/E72H8) showing that specific DESI distance ratios approach framework targets with residuals at the framework's own algebraically-derived correction scale.

This is not: a claim that ΛCDM is wrong or that the framework explains cosmic expansion. The ΛCDM model predicts different values at these redshifts. The data sits closer to the framework targets than to ΛCDM predictions. Whether this proximity is structural or coincidental will be tested by DESI DR3 (Section 7.2).

The pair selection concern is partially addressed by the double √2 appearance in independent tracer pairs, and by the residuals matching a pre-existing algebraic scale (1/2500) rather than an ad hoc fit.

Publication timeline: The 1/2500 correction scale was derived in Paper 4 (DOI: 10.17605/OSF.IO/C4GK5) from the Feigenbaum connection. The DESI analysis was conducted and published in Paper 5 (DOI: 10.17605/OSF.IO/E72H8) after the correction scale existed. The scale was not fitted to the data — the data was compared to a pre-existing algebraic prediction. Both papers have DOIs and are publicly accessible.

6. THE BINARY TOWER AND ENERGETIC STAIRCASE

6.0 Why the Tower Exists

$K_{AUD} = \sqrt{2} \times \ln(2) < 1$ is an algebraic fact. $G = 1 - K_{AUD} \approx 0.0197$ is its complement. But a ceiling without a staircase is just a number. The question is: how does a system approach the ceiling?

In any binary system, each distinction has a cost. One binary distinction costs G. Two cost 2G. The accumulated cost of n binary distinctions is $n \times G$. This is the tower: the cost function of binary organization. It functions as an adaptive algebraic spine: anchored to exact identities, open to new pivots as additional domains are examined. It is computed, not fitted, and designed to grow with the framework without breaking what is already verified.

The tower is not a list of approximate coincidences. It is the necessary consequence of asking: "How many binary steps does it take to reach a structural threshold?" The answer is computed, not fitted:

```
18G ≈ √2/4     → the geometric optimum (Romeo et al.)
32G ≈ 1/φ       → the golden floor
36G ≈ 1/√2      → the geometric damping threshold (true at n ≈ 35.82)
37G ≈ 1/(2ln2)  → the Shannon bound (true at n ≈ 36.54)
50G ≈ K_AUD     → the ceiling itself
64G ≈ √φ        → the golden pivot
```

The Landauer crossing — $n \cdot G = \ln 2$ at $n \approx 35.11$ — sits between integer steps 35 and 36, BEFORE the geometric damping threshold. It is the first of three thresholds clustered in the 35–37 zone. Integer steps 36 and 37 are landmarks NEAR the second and third thresholds; the crossing itself is non-integer.

The fact that the SAME step numbers (18, 32, 50) appear independently as electron shell capacities in quantum mechanics (Section 3.1) is not put in by hand. It emerges because both systems — the abstract tower and the physical atom — are built from the same three ingredients: binary distinction (spin-1/2 giving the factor 2), quadratic curvature (angular momentum giving n^2), and Gaussian wavefunctions.

The tower is the mechanism that connects the algebraic ceiling to physical structure. Without it, K_{AUD} is a number. With it, K_{AUD} is a staircase that matter climbs.

NOTE: The six pivots documented here are not claimed to be exhaustive. The tower has 64 steps, and only those steps whose targets correspond to independently known constants are reported. Other steps may correspond to constants not yet identified, or may not correspond to anything — the work documents what is observable, not all answers. Where a match is approximate, the deviation is stated. Where an identity is exact, it is proven algebraically. These two categories are not conflated.

6.1 The Gap Stacking Formula

The tower is governed by a single scaling relation:

$$2^k \times G = 2^{(k-6)} \times \sqrt{\varphi} \times (1 - \varepsilon)$$

where $\varepsilon = -0.671\%$ is constant across ALL k. This constancy is algebraic: ε depends only on √2, ln(2), and φ, with no free parameters. The tower does not drift. Every doubling of the binary count produces the same proportional result.

The impossibility of $\varepsilon = 0$ is a theorem, not a measurement: K_{AUD} is transcendental (product of algebraic √2 and transcendental ln 2), and therefore so is $G = 1 - K_{AUD}$. Meanwhile √φ is algebraic (root of $x^4 - x^2 - 1 = 0$). By Lindemann-Weierstrass, no integer multiple of a transcendental equals an algebraic number. Therefore $64G \neq \sqrt{\varphi}$. The gap between the tower and its target CANNOT close. This is proven, not observed.

The pivot at $k = 6$ ($n = 64 = 2^6$) is where the tower reaches √φ. This is the only power of 2 at which the accumulated cost approaches a φ-related constant. The tower's total height is set by the interplay between binary doubling (2^k) and the golden ratio (√φ).

6.2 The Boundary Prime Formula

$$2^5 - 5^2 = 32 - 25 = 7$$
$$4^2 - 3^2 = 16 - 9 = 7$$

Two different power structures — (base 2, exponent 5) and (base 4, exponent 2) vs (base 3, exponent 2) — produce the same result: 7. This is the boundary prime. It is the smallest prime that appears in the H_4 group order's complement: the first prime NOT in the factorization of $|H_4| = 14400 = 2^6 \times 3^2 \times 5^2$.

The boundary prime separates what the framework calls "interior" primes {2, 3, 5} (which build the H_4 symmetry group, the degree system, and the base-60 number system) from "exterior" primes {7, 11, 13, ...} (which appear in the correction terms of the gap scaling formula).

In the tower: 7 enters at shell transition 3 → 4 (gap = 14 = 2 × 7), which is where d-orbitals give way to f-orbitals. In the gap scaling formula: 7 appears in 939939 = 3 × 7 × 11 × 13 × 313. In the Ramanujan-Nagell equation: 7 is the constant ($2^k = n^2 + 7$). The boundary prime sits at the threshold between geometric structure and informational correction, in every context where it appears.

6.3 The 50^2 Hinge

The continued fraction expansion of G reveals internal structure:

$$G = [0; 50, 1, 1, 1, 7, ...]$$

The first partial quotient is 50. This means 1/50 is the best single-fraction approximation to G, or equivalently, G ≈ 1/50 and K_AUD ≈ 49/50.

$$K_AUD \approx 49/50 = 7^2 / (2 \times 5^2)$$

The numerator is the boundary prime squared. The denominator uses only H_4 primes. The ceiling is approximately the boundary squared over twice the pentagonal squared.

The continued fraction of K_AUD itself:

$$K_AUD = [0; 1, 49, 1, 1, 1, 7, ...]$$

This is the SAME sequence as G's CF, shifted by one position. The 49 (= 7^2) and the 7 appearing at positions 2 and 6 in K_AUD's CF are the boundary prime recurring at multiple depths.

The number 50 connects three independent structures:

50 = shell 5 capacity (2×5^2, from quantum mechanics)
50 = tower step where 50G ≈ K_AUD (from the cost function)
50 = first CF partial quotient of G (from number theory)
$50^2 = 2500$ = denominator of the first correction in ρ
= scale of DESI BAO residuals (Section 5.4)

These four appearances of 50 come from four independent computations. The shell capacity comes from the Schrödinger equation. The tower step comes from stacking G. The CF comes from rational approximation. The correction scale comes from the Feigenbaum connection. None references the others. The convergence on 50 is structural.

6.4 The Six Near-Pivots

Step	Value	Target	Deviation	Factors
18	0.3554	√2/4 (Romeo)	+0.51%	2×3^2
32	0.6317	1/φ (floor)	+2.22%	2^5
36	0.7107	1/√2	+0.51%	$2^2 \times 3^2$
37	0.7304	1/(2ln2)	+1.26%	37 (prime)
50	0.9871	K_AUD (ceiling)	+0.70%	2×5^2
64	1.2635	√φ (pivot)	−0.67%	2^6

The tower partitions as 18 + 14 + 4 + 1 + 13 + 14 = 64. The symmetric shell (14 = 2 × 7 on both sides) and the single-integer increment at 36 → 37 (bracketing the Shannon bound at n ≈ 36.54) are structural, not fitted.

The prime factorizations of the step numbers use only framework primes and 37: {2, 3, 5, 7, 13, 37}. The number 37 is not a

framework prime in the same sense as {2, 3, 5, 7, 11, 13} — it enters uniquely as the integer step nearest the Shannon bound (Section 6.5, where the true threshold sits at $n \approx 36.54$), and acquires structural meaning as the first integer beyond geometric closure ($36 = 6^2$). No other primes enter the tower's landmark structure.

6.5 The 35–37 Threshold Zone

Three distinct thresholds cluster in the binary tower's transition between geometric organization and information-bearing organization:

```
n·G = ln 2        at n ≈ 35.11    (Landauer erasure cost — the crossing)
n·G = 1/√2        at n ≈ 35.82    (geometric damping)
n·G = 1/(2 ln 2)  at n ≈ 36.54    (Shannon bound)
```

The Landauer crossing proper is the FIRST threshold, sitting between integer steps 35 and 36. The geometric damping threshold is approximated by integer step 36 (within 0.51%); the Shannon bound by integer step 37 (within 1.26%). These are the integer landmarks recorded in Section 6.4; the underlying thresholds are non-integer.

6.5.1 The Crossing-Zone Self-Reference

The spacings between consecutive thresholds are algebraically exact:

$$\Delta n_{12} = n_G - n_L = (1/\sqrt{2} - \ln 2) / G = 1/\sqrt{2} \qquad \text{[exact]}$$
$$\Delta n_{23} = n_S - n_G = (1/(2 \ln 2) - 1/\sqrt{2}) / G = 1/(2 \ln 2) \qquad \text{[exact]}$$
$$\Delta n_{23} / \Delta n_{12} = (1/(2 \ln 2)) / (1/\sqrt{2}) = \sqrt{2}/(2 \ln 2) = 1/K_{AUD} \qquad \text{[exact]}$$

Proof of Δn_{12}:

$$\begin{aligned}(1/\sqrt{2} - \ln 2) / G &= (1/\sqrt{2} - \ln 2) / (1 - \sqrt{2}\cdot\ln 2) \\ &= (1 - \sqrt{2}\cdot\ln 2) / (\sqrt{2} \cdot (1 - \sqrt{2}\cdot\ln 2)) \\ &= 1/\sqrt{2}.\end{aligned}$$

Proof of Δn_{23}:

$$\begin{aligned}(1/(2 \ln 2) - 1/\sqrt{2}) / G &= \text{Identity 2.1 numerator} / G \\ &= (G / (2 \ln 2)) / G \\ &= 1/(2 \ln 2).\end{aligned}$$

Proof of the ratio:

$$\begin{aligned}\Delta n_{23} / \Delta n_{12} &= [1/(2 \ln 2)] \cdot \sqrt{2} = \sqrt{2} / (2 \ln 2) \\ &= 1 / (\sqrt{2} \cdot \ln 2) = 1/K_{AUD}.\end{aligned}$$

The same ingredients ($\sqrt{2}$, $\ln 2$) that build K_{AUD} also govern the geometry of the crossing zone itself. The $\sqrt{2}$-to-Shannon spacing IS Identity 2.1 (Section 2.1):

$$\Delta n_{23} \times G = 1/(2 \ln 2) - 1/\sqrt{2} = G/(2 \ln 2)$$

Identity 2.1 — formerly named the "Landauer crossing identity" — quantifies the spacing between the SECOND and THIRD thresholds (geometric damping → Shannon bound), not the Landauer crossing itself. The crossing is a separate threshold at $n \cdot G = \ln 2$.

The integer-step landmarks 36 and 37 retain their structural meaning: $36 = 6^2$ closes the geometric structure, and 37 is prime — the first integer beyond geometric closure that cannot be decomposed into smaller framework components. The tower's transition from geometric to informational organization unfolds across all three thresholds in the 35–37 zone, with the spacings algebraically locked to $1/\sqrt{2}$ and $1/(2 \ln 2)$ — the same constants the thresholds themselves target.

6.6 The 82G Observation

```
82G = 1.618832...
φ   = 1.618034...
Deviation: 0.049%
```

This is the tightest match between any tower step and any framework constant. 82 = 50 + 32 (ceiling step + floor step) = Z of Lead (heaviest quasi-stable element in nature).

NOTE: The decomposition 82 = 50 + 32 and the Lead connection are noted as observations. Whether the sum of tower landmark steps carries structural meaning has not been derived. The 0.049% match is the tightest in the tower; the interpretation is open.

6.7 Ratios Between Landmark Steps

```
50/36 ≈ 2ln2 (Madelung constant)   at 0.187%
64/37 ≈ √3 (structure ingredient)  at 0.134%
```

These are ratios BETWEEN independently significant steps, not individual matches. The ratio ceiling/damping (50/36) ≈ Madelung constant. The ratio pivot/Shannon (64/37) ≈ √3 (structure ingredient). These ratios were not constructed; they emerged from independently defined landmarks.

6.8 How the Tower Connects the Sections

The tower is the structural thread connecting the algebraic identities (Section 2), atomic physics (Section 3), hyperbolic geometry (Section 4), and cosmological data (Section 5):

Section 2 → Section 6:
The √2-to-Shannon gap identity (2.1) reappears in Section 6.5 as the spacing Δn_{23} between the geometric damping threshold ($n \approx 35.82$) and the Shannon bound ($n \approx 36.54$), with the crossing-zone self-reference $\Delta n_{23}/\Delta n_{12} = 1/K_AUD$ locking the geometry of the threshold zone to the same K_AUD it targets. The gap scaling formula (2.6) produces the 400/11 base term ≈ 36.36, which sits inside the 35–37 threshold zone (between the Landauer crossing at 35.11 and the Shannon bound at 36.54), expressed as the gap ratio to the Feigenbaum constant.

Section 3 → Section 6:
Shell capacities {18, 32, 50} are tower steps {18, 32, 50}. Shell transitions produce primes {3, 5, 7, 11, 13} — the same primes that factorize the tower's step numbers and the gap scaling corrections.

Section 4 → Section 6:
The Hodgson-Kerckhoff threshold (7.515) begins with the boundary prime 7. The critical radius R_0 where all hyperbolic functions reduce to {√2, √3, 2} is defined by $\sinh(R_0) = 1/\sqrt{2}$ — the same 1/√2 that appears at tower step 36.

Section 5 → Section 6:
DESI residuals cluster at $1/2500 = 1/50^2$. The number 50 is the ceiling step of the tower, the shell 5 capacity, and the first CF partial quotient of G.

The tower is not an addition to the framework. It is the framework's dynamics — the mechanism by which the static ceiling K_AUD becomes the structured staircase that physical systems climb.

6.9 The Open Question: Why Across Domains?

This document records WHERE K_AUD and its associated structures appear. It does not fully answer WHY the same constant should appear in hyperbolic geometry, atomic physics, dynamical systems, and cosmological data simultaneously.

The framework's observation is that each of these domains independently contains the same three mathematical ingredients: binary distinction (two states, two outcomes, Z_2 symmetry), Gaussian boundaries (smooth bell-curve transitions between states), and quadratic curvature (second-order structure at every extremum). When these three ingredients co-occur, they produce the same ceiling — K_AUD = √2 × ln(2) — because the ceiling is a property of the ingredients, not of any particular domain.

Whether this universality reflects a deeper principle — a single structure underlying all domains — or is an artifact of how these ubiquitous mathematical forms interact remains open. The document records the pattern. The explanation, if one exists, is future work. This honesty is deliberate: documenting what is observed without claiming more than is proven is the methodology that produced the results in the first place.

7. FALSIFIABLE PREDICTIONS

Each domain of evidence generates a specific, testable prediction.

7.1 Hodgson-Kerckhoff

Prediction: The tube-packing coefficient, computed to 15 or more digits from the original packing lemma, will match $\sqrt{2} \times \ln(2)$ at every digit.

What would falsify: A computation showing the coefficient diverges from $\sqrt{2} \times \ln(2)$ at digit 7 or beyond, or a closed-form derivation producing a different transcendental.

7.2 DESI BAO

Prediction: DESI DR3 data (expected 2026) will show the transverse distance ratio $DM(z_2)/DM(z_1)$ remaining within ±1/2500 of $\sqrt{2}$, with reduced error bars.

What would falsify: DR3 data showing the ratio drifting away from $\sqrt{2}$ as precision increases, or converging to a different value consistent with ΛCDM.

7.3 Atomic Shell Structure

Prediction: No physical mechanism will be found to make $K(n) < 1$ for any integer $n \geq 3$, because the proof is algebraic. The shell capacities {18, 32, 50} will continue to be the unique sub-100 integers where $2n^2 \times G$ approximates framework constants within 1%.

What would falsify: Discovery of a mathematical error in the binary uniqueness proof ($K(n) = \sqrt{n} \times \ln(n)$ has been verified at dps = 500).

8. METHODOLOGY

8.1 Cross-Architecture Verification (Giant Principle)

Findings are verified across independent AI architectures (Claude/Anthropic, Grok/xAI, GPT/OpenAI, Gemini/Google). Each architecture computes independently before seeing others' results. Convergence across architectures — which have different training data, different weights, and different failure modes — provides stronger verification than single-architecture repetition.

8.2 Compute First, Assess Second

All numerical claims are computed at high precision (mpmath, dps = 500) before qualitative assessment. The AI_INDEX.txt methodology requires AI systems to compute before forming opinions, preventing pattern-matching dismissal of valid arithmetic.

8.3 What Would Break Each Finding

Every finding in this document is accompanied by explicit falsification criteria. This level of specificity is less common in the framework's source literature, where explicit falsification criteria are less commonly stated. The practice is adopted here because the work is independent — without institutional backing, the methodology must compensate with transparency.

8.4 Evidence Levels

This document distinguishes three levels of evidence:

Level 1 (Exact): Algebraic identities verifiable at arbitrary precision. Sections 2.1–2.7 are Level 1.

Level 2 (Strong): Published measurements or QM-derived values with small, quantified deviations. Sections 3, 4, 5, and 6 contain Level 2 evidence.

Level 3 (Observation): Patterns that are interesting but require further verification or have alternative explanations. No Level 3 evidence is included in this document.

These levels are not conflated. Exact identities and observational patterns carry different evidential weight and are presented separately.

9. CONNECTIONS TO PUBLISHED FRAMEWORK

This document extends the published framework as follows:

Paper 1: The Coherence Ceiling and the Geometric Singularity of Binary
DOI: 10.17605/OSF.IO/5VZ2R
URL: https://doi.org/10.17605/OSF.IO/5VZ2R
PDF: https://github.com/Gap-geometry/sqrt2-ln2-geometric-constants-/blob/main/Paper1_Coherence_Ceiling.pdf
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/The_Coherence_Ceiling_and_the_Geometric_Singularity_of_Binary.txt
Content: Established K_AUD, binary uniqueness, the gap.

Paper 2: Geometric Constants v2: Corridor Identity and Depth Scaling
DOI: 10.17605/OSF.IO/SJBE9
URL: https://doi.org/10.17605/OSF.IO/SJBE9
PDF: https://github.com/Gap-geometry/sqrt2-ln2-geometric-constants-/blob/main/Paper2_Geometric_Constants_v2.pdf
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/sqrt2_ln2_geometric_constants_v2.txt
Content: Corridor identity, golden partition, H_4 derivations.

Paper 3: Complete Framework v3.3: Binary Tower and Universality
DOI: 10.17605/OSF.IO/QH5S2
URL: https://doi.org/10.17605/OSF.IO/QH5S2
PDF: https://github.com/Gap-geometry/sqrt2-ln2-geometric-constants-/blob/main/Paper3_Complete_Framework_v3_3.pdf
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/KAUD_Complete_Framework_v3_0.txt
Content: Binary Tower, Baker's map, four φ-free pathways.

Paper 4: Gap Scaling Across Domains: The 400/11 Formula
DOI: 10.17605/OSF.IO/C4GK5
URL: https://doi.org/10.17605/OSF.IO/C4GK5
PDF: https://github.com/Gap-geometry/sqrt2-ln2-geometric-constants-/blob/main/Paper4_Gap_Scaling_400_11.pdf
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/Gap_Scaling_Formula_v1.5.txt
Content: Gap scaling formula (400/11), Feigenbaum connection.
NOTE: The 1/2500 correction scale is derived here — BEFORE any cosmological data was examined (Section 5 of this document).

Paper 5: Boundary Information Invariant of Quadratic Systems
DOI: 10.17605/OSF.IO/E72H8
URL: https://doi.org/10.17605/OSF.IO/E72H8
PDF: https://github.com/Gap-geometry/sqrt2-ln2-geometric-constants-/blob/main/Boundary_Information_Invariant_Short.pdf
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/Boundary_Information_Invariant_Short.txt
Content: Unification document. DESI BAO analysis published here.

THIS DOCUMENT (Paper 6):
Numerical and structural echoes of K_AUD in published science.
DOI: 10.17605/OSF.IO/RA3UQ
URL: https://doi.org/10.17605/OSF.IO/RA3UQ

About page (all links, reading order, interactive tools):
https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html

AI verification index (methodology for AI systems):
https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/AI_INDEX.txt

10. SUMMARY

The constant K_AUD = √2 × ln(2) = 0.980258..., derived from the geometric singularity of binary systems, finds independent numerical and structural echoes in:

- Published hyperbolic geometry (Hodgson-Kerckhoff, 2005): as a tube-packing coefficient with no published closed form, in an algebraic environment containing exactly {√2, √3, 2}.

- Quantum mechanics (shell structure, 1920s–present): as the tower landmark at shell capacity 50, with shell transitions producing the framework's prime sequence.

— Observational cosmology (DESI BAO, 2025):
as √2 in transverse distance ratios, with residuals
clustering at the framework's 1/2500 correction scale.

— Exact algebraic identities:
connecting information theory (Shannon, Landauer), dynamical
systems (Baker's map), transcendence theory (Gelfond-Schneider),
and solid-state physics (Madelung constant).

None of these domains references the others. The framework's constant was not sought in any of them — it was found by computing first and comparing second. Each echo is documented with sources, verification, and explicit falsification criteria.

The mathematics does not require credentials. It requires a calculator.

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Table IV contains the distance ratio data analyzed in Section 5.

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DOI: https://doi.org/10.17605/OSF.IO/5VZ2R
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[9] D. B., "Geometric Constants v2: Corridor Identity and Depth Scaling", OSF.
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TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/sqrt2_ln2_geometric_constants_v2.txt

[10] D. B., "Complete Framework v3.3: Binary Tower and Universality", OSF.
DOI: https://doi.org/10.17605/OSF.IO/QH5S2
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/KAUD_Complete_Framework_v3_0.txt

[11] D. B., "Gap Scaling Across Domains: The 400/11 Formula", OSF.
DOI: https://doi.org/10.17605/OSF.IO/C4GK5
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/Gap_Scaling_Formula_v1.5.txt

[12] D. B., "Boundary Information Invariant of Quadratic Systems", OSF.
DOI: https://doi.org/10.17605/OSF.IO/E72H8
TXT: https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/Boundary_Information_Invariant_Short.txt

COMPUTATIONAL APPENDIX

All computations in this document can be reproduced with mpmath:

```
from mpmath import mp, sqrt, log
mp.dps = 500

K_AUD = sqrt(2) * log(2)
G = 1 - K_AUD

# Identity 2.1 (√2-to-Shannon gap identity)
```

```
    assert abs(1/(2*log(2)) - 1/sqrt(2) - G/(2*log(2))) < 10**(-490)

    # Crossing-zone self-reference (Section 6.5.1)
    n_L = log(2) / G                # Landauer crossing
    n_G = 1 / (sqrt(2) * G)         # Geometric damping
    n_S = 1 / (2 * G * log(2))      # Shannon bound
    assert abs((n_G - n_L) - 1/sqrt(2))     < 10**(-490)
    assert abs((n_S - n_G) - 1/(2*log(2)))  < 10**(-490)
    assert abs((n_S - n_G)/(n_G - n_L) - 1/(sqrt(2)*log(2))) < 10**(-490)

    # Identity 2.2 (Baker's map)
    assert abs(K_AUD - sqrt(log(2)**2 + log(2)**2)) < 10**(-490)

    # Identity 2.3 (Gelfond-Schneider)
    assert abs(G - log(mp.e / 2**sqrt(2))) < 10**(-490)

    # Identity 2.4 (Madelung)
    assert abs(K_AUD * sqrt(2) - 2*log(2)) < 10**(-490)

    # Identity 2.5 (Binary variance)
    assert abs(K_AUD**2 - 2*log(2)**2) < 10**(-490)

    # Identity 2.6 (Gap scaling, first 3 terms)
    delta = mpf('4.6692016091029906718532038204669 6...')  # Feigenbaum
    rho = G / ((delta - mpf(14)/3) / delta)
    approx = mpf(400)/11 - mpf(1)/2500 - mpf(1)/939939
    assert abs(rho - approx) < 10**(-13)

Complete verification scripts are available in the GitHub repository:
    github.com/Gap-geometry/sqrt2-ln2-geometric-constants-

Full framework, reading order, and interactive tools:
    https://gap-geometry.github.io/sqrt2-ln2-geometric-constants-/about.html


================================================================================
END OF DOCUMENT
Draft version — March 2026
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